

Tubulointerstitial — Autoimmune AIN gallery



1. Autoimmune causes list

WHAT YOU SEE

A two-column museum wall placard (Absolute / Relative) listing Sjögren, SLE, TINU, IgG4, sarcoid, granulomatous, post-infectious — the gallery's roster of non-drug, autoimmune AIN causes.

WHAT IT ENCODES

Beyond drugs, acute interstitial nephritis has a systemic/autoimmune bucket: Sjögren, SLE, TINU (tubulointerstitial nephritis + uveitis), IgG4-related disease, sarcoidosis (granulomatous), and post-infectious causes. These share a T-cell-mediated interstitial infiltrate and, unlike most drug AIN, are often steroid-responsive and may need a maintenance immunosuppressant (e.g., azathioprine, MMF) to prevent relapse. Mnemonic: 'Some Systemic Things Inflamm Glomeruli's neighbors Severely' — Sjögren, SLE, TINU, IgG4, Granulomatous/sarcoid.

BOARD TRAP

A young patient with AIN, normal urine culture and no drug exposure is given 'idiopathic AIN, observe.' Wrong — absent a culprit drug, hunt for a systemic cause (Sjögren antibodies, ACE/calcium for sarcoid, IgG4 level, slit-lamp for uveitis); these autoimmune forms benefit from steroids, so 'just observe' loses treatable disease.

2. TINU

WHAT YOU SEE

Young patient on scale, fever, eye 'uveitis' clue, statue pedestal TINU

WHAT IT ENCODES

TINU = Tubulointerstitial Nephritis and Uveitis, a steroid-responsive syndrome typically in adolescent girls/young women (median early-to-mid teens) but also adults. Triad: AIN (sterile pyuria, tubular dysfunction, AKI), bilateral anterior uveitis (often follows the renal disease and recurs), and systemic symptoms — fever, weight loss, fatigue, anemia, elevated ESR. Urinary β2-microglobulin is markedly elevated; treat with glucocorticoids and refer to ophthalmology because uveitis can be sight-threatening.

BOARD TRAP

A teen with AIN and red painful eyes is diagnosed with 'sarcoid uveitis' or simple 'conjunctivitis.' Wrong — sarcoid would show hypercalcemia, hilar adenopathy and non-caseating granulomas; TINU is granuloma-light with anterior uveitis and a normal chest film. Do not restrict TINU to adults — it is classically a young-female disease, and the eye exam (anterior uveitis) is the discriminator.

3. Sjogren

WHAT YOU SEE

Veiled statue with wolf, pedestal SJOGREN / TYPE I RTA

WHAT IT ENCODES

Sjögren syndrome causes a lymphocytic tubulointerstitial nephritis whose signature is DISTAL (type I) RTA — the collecting-duct H⁺-ATPase is targeted, giving hypokalemic hyperchloremic non-gap acidosis, inappropriately alkaline urine (pH >5.5), nephrocalcinosis and stones. Patients have sicca (dry eyes/mouth), positive anti-Ro/SSA and anti-La/SSB. Severe hypokalemia can cause periodic paralysis as the presenting event.

BOARD TRAP

Sjögren renal disease is paired with 'type IV (hyperkalemic) RTA.' Wrong — Sjögren classically causes type I DISTAL RTA with HYPOkalemia and inability to acidify urine; type IV (hyperkalemic) is the diabetic/hypoaldosterone pattern. The potassium direction (low in Sjögren type I) is the discriminator.

4. SLE / lupus

WHAT YOU SEE

A statue holding a butterfly-rash (malar) mask on the pedestal 'SLE / TYPE IV RTA' — lupus, whose tubulointerstitial hit rides alongside its glomerular nephritis.

WHAT IT ENCODES

Lupus most famously causes glomerular (proliferative/membranous) nephritis, but tubulointerstitial involvement independently worsens prognosis and can produce distal or type IV RTA and tubular dysfunction. The degree of tubulointerstitial inflammation/fibrosis on lupus nephritis biopsy predicts progression to CKD as strongly as the glomerular class. Look for low complements, anti-dsDNA, and multisystem disease.

BOARD TRAP

In an SLE patient with rising creatinine, the answer assumes 'pure tubulointerstitial nephritis, no glomerular disease, skip the biopsy.' Wrong — SLE renal involvement is dominated by glomerular lupus nephritis and biopsy is needed to classify it (class III/IV/V) and guide induction immunosuppression; the tubulointerstitial component coexists rather than replaces it.

5. Granulomatous / sarcoid

WHAT YOU SEE

A grey stone giant studded with face-granulomas, holding a snowflake (hypercalcemia) and a chest X-ray, pedestal 'GRANULOMATOUS' with 'RULE OUT TB FIRST' — sarcoid granulomas plus 1,25-vitamin-D-driven hypercalcemia.

WHAT IT ENCODES

Sarcoidosis causes granulomatous (non-caseating) interstitial nephritis, but the more common renal hit is from hypercalcemia/hypercalciuria — activated macrophages make 1-alpha-hydroxylase converting 25-OH-D to active 1,25-(OH)₂-D, driving gut calcium absorption, nephrocalcinosis, stones and a concentrating defect. Look for hilar adenopathy, elevated ACE, and a brisk steroid response.

BOARD TRAP

Hypercalcemia in a sarcoid patient is treated with 'IV calcitriol/vitamin D' or worked up as primary hyperparathyroidism. Wrong — sarcoid hypercalcemia is from EXCESS 1,25-vitamin D with a SUPPRESSED PTH; giving vitamin D worsens it. Treatment is glucocorticoids (and sun/dietary calcium avoidance); the suppressed PTH with high 1,25-D is the discriminator from primary HPT.

6. Steroids work

WHAT YOU SEE

A gold-starred 'STEROIDS' archway with a green-checked 'STEROIDS WORK' sign over a crossed-out 'PSEUDOTUMOR' — the gold star marks glucocorticoids as the answer for these autoimmune/granulomatous AINs.

WHAT IT ENCODES

The autoimmune/granulomatous AINs (TINU, sarcoid, IgG4, lupus, Sjögren) are characteristically glucocorticoid-responsive — prednisone ~1 mg/kg/day (or pulse methylprednisolone in severe disease) with a taper; IgG4 and TINU may need steroid-sparing maintenance (azathioprine, MMF) to prevent relapse. This contrasts with chronic toxin/structural CIN, where fibrosis is fixed and steroids do not help. In drug AIN, the first move is always stopping the drug, with steroids reserved if creatinine does not improve.

BOARD TRAP

Glucocorticoids are recommended for a patient with CHRONIC interstitial fibrosis (e.g., chronic lithium or analgesic nephropathy). Wrong — established fibrosis is irreversible and steroids do not help; steroid responsiveness applies to acute cellular autoimmune AIN, not chronic scarred CIN. Acuity/cellular infiltrate vs fibrosis on biopsy is the discriminator.

7. IgG4 disease

WHAT YOU SEE

Pancreas/organ figure, IgG4 pedestal, pseudotumor

WHAT IT ENCODES

IgG4-related disease is a fibroinflammatory disorder with IgG4-plasma-cell-rich storiform fibrosis that affects pancreas (autoimmune pancreatitis), salivary glands, retroperitoneum, and kidney (IgG4 tubulointerstitial nephritis). Renal involvement may form a mass-like 'pseudotumor' or low-density cortical lesions mimicking malignancy or lymphoma; serum IgG4 is often elevated and complements may be low. Diagnosis is by biopsy (storiform fibrosis, >40% IgG4+/IgG plasma cells, obliterative phlebitis); treat with steroids ± rituximab.

BOARD TRAP

An IgG4 renal pseudotumor is taken to the OR as presumed 'renal cell carcinoma / lymphoma' for nephrectomy. Wrong — IgG4 disease mimics tumor but is a steroid-responsive inflammatory mass; checking serum IgG4, looking for synchronous organ involvement (pancreas, salivary), and biopsy avoid an unnecessary nephrectomy. Multi-organ involvement + high IgG4 is the discriminator from true malignancy.

8. PD-1 robot (drug overlap)

WHAT YOU SEE

Robot labeled PD-1 entering gallery

WHAT IT ENCODES

Immune checkpoint inhibitors (anti–PD-1 nivolumab/pembrolizumab, anti–PD-L1, anti–CTLA-4 ipilimumab) unleash T cells and are now a leading cause of AIN in oncology patients — a drug-induced but immune-mediated lesion that bridges the 'drug' and 'autoimmune' buckets. AKI typically appears weeks-to-months into therapy, often with pyuria and other immune-related adverse events (colitis, dermatitis, thyroiditis). Concurrent PPIs/NSAIDs raise risk; management is hold the inhibitor and give glucocorticoids.

BOARD TRAP

AKI in a patient on pembrolizumab is reflexively called 'contrast nephropathy' or 'prerenal/tumor-related.' Wrong — checkpoint-inhibitor AIN is the high-yield cause, is steroid-responsive, and warrants holding the drug; missing it forfeits a treatable lesion. The temporal link to immunotherapy plus pyuria and other immune-related toxicities is the discriminator.

9. Fever + uveitis + weight loss (TINU)

WHAT YOU SEE

Boxed clues: fever, uveitis, weight loss, age 2-15

WHAT IT ENCODES

The TINU clue cluster: fever, anterior UVEITIS (eye redness/pain/photophobia), unexplained weight loss and fatigue, plus AKI with sterile pyuria — most often in girls/young adults. Markedly elevated urinary β2-microglobulin and a high ESR support it. The uveitis may precede, accompany, or follow the nephritis and tends to recur, so ophthalmology follow-up is mandatory even after the kidney recovers.

BOARD TRAP

A young patient with AKI, fever and eye symptoms is diagnosed with 'Behçet disease' or 'reactive arthritis.' Wrong — those have distinct features (oral/genital ulcers and pathergy in Behçet; urethritis/arthritis triad in reactive arthritis); the renal AIN with anterior uveitis and high urinary β2-microglobulin points specifically to TINU. The kidney biopsy showing AIN is the discriminator.

10. Pyelo ≠ AIN

WHAT YOU SEE

Bottom sign 'PYELO ≠ AIN'

WHAT IT ENCODES

Acute pyelonephritis is an INFECTIOUS tubulointerstitial inflammation (ascending bacteria, positive culture, WBC casts, flank pain, fever) and is treated with antibiotics — distinct from the STERILE, immune/drug-mediated AIN. Both can show pyuria and WBC casts, so the discriminator is the urine culture: positive → pyelonephritis; negative (sterile pyuria) with a drug/systemic trigger → AIN.

BOARD TRAP

Sterile pyuria with WBC casts after starting a new antibiotic is treated as 'breakthrough pyelonephritis, broaden antibiotics.' Wrong — a negative culture with a temporally linked drug indicates drug-induced AIN; escalating the (causative) antibiotic worsens it. The negative culture is the key discriminator from true pyelonephritis.

11. BK reactivation (transplant)

WHAT YOU SEE

Transplant/crystal-ball figure, 'BK REACTIVATION'

WHAT IT ENCODES

BK polyomavirus nephropathy is an interstitial nephritis from viral reactivation in the over-immunosuppressed kidney transplant recipient (usually 3–12 months post-transplant). It mimics rejection (rising creatinine) but the treatment is OPPOSITE — REDUCE immunosuppression rather than augment it. Diagnose with plasma BK PCR (rising viral load) and biopsy showing viral cytopathic 'decoy cells' and intranuclear inclusions; urine decoy cells are a screening clue.

BOARD TRAP

A transplant patient with rising creatinine and BK viremia is given 'pulse steroids / increased immunosuppression for acute rejection.' Wrong — that worsens BK nephropathy; the correct move is to REDUCE immunosuppression and monitor BK PCR. Biopsy (SV40 staining for BK vs tubulitis/C4d for rejection) is the discriminator, and the management direction is reversed.

12. Post-COVID / post-infectious kids

WHAT YOU SEE

Child with rash, 'POST-COVID KIDS'

WHAT IT ENCODES

Post-infectious interstitial nephritis (classically after streptococcal, leptospirosis, Legionella, and now SARS-CoV-2/post-COVID) is an immune-mediated AIN seen notably in children, sometimes with a TINU-like picture. It follows an infection by days–weeks, gives AKI with sterile pyuria and tubular dysfunction, and is generally self-limited or steroid-responsive. It is distinct from acute infection of the kidney itself.

BOARD TRAP

A child with AKI after a recent viral illness is diagnosed with 'post-streptococcal glomerulonephritis.' Wrong — PSGN gives a nephritic picture with RBC casts, hematuria, hypertension, edema and LOW complement (C3); post-infectious interstitial nephritis gives sterile pyuria, tubular proteinuria and normal complement. The sediment (RBC casts/low C3 vs sterile pyuria/normal C3) is the discriminator.